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$$f(x, y) = (x^2 - y^2, 2xy)$$

$$Df(x, y) = \begin{pmatrix} \frac{\partial}{\partial x}(x^2 - y^2) & \frac{\partial}{\partial y}(x^2 - y^2) \\ \frac{\partial}{\partial x}(2xy) & \frac{\partial}{\partial y}(2xy) \end{pmatrix}$$
$$= \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}$$

$$\det(Df(x, y)) = (2x)(2x) - (-2y)(2y) = 4x^2 + 4y^2 = 4(x^2 + y^2)$$

$$\Rightarrow \det(Df(x, y)) \neq 0, \forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$$

$$\Rightarrow (Df^{-1})(f(x, y)) = (Df(x, y))^{-1} = \frac{1}{4(x^2 + y^2)} \begin{pmatrix} 2x & 2y \\ -2y & 2x \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2x}{4(x^2 + y^2)} & \frac{2y}{4(x^2 + y^2)} \\ \frac{-2y}{4(x^2 + y^2)} & \frac{2x}{4(x^2 + y^2)} \end{pmatrix}$$

References